Dimension of Subspaces and the Rank-Nullity Theorem

Lecture Notes

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Introduction

This lecture connects several core concepts: subspaces, basis, and dimension. A **subspace** is a vector space contained within a larger one. A **basis** is a minimal set of vectors that describes the entire subspace. The **dimension** is the number of vectors in its basis, giving a precise measure of its "size."

We will learn to find the dimension of various subspaces, including the three fundamental subspaces of a matrix: the **Null Space**, **Column Space**, and **Row Space**. Finally, we will see how these dimensions are related by the **Rank-Nullity Theorem**.

1 Finding the Dimension of a General Subspace

Key Idea

The **dimension** of a subspace is the number of vectors in any basis for that subspace. The most direct way to find the dimension is to first find a basis. A common method is to "factor out" the free parameters from a general vector form to reveal the basis vectors.

1.1 Example 1: A Subspace of $\mathbb{R}^3$

Problem: Find a basis and the dimension of the subspace $U \subset \mathbb{R}^3$ of all vectors of the form $\begin{bmatrix} x+y \\ y \\ -3x \end{bmatrix}$.

Solution (Step-by-Step)

1. **Decompose the General Vector:** Separate by parameter.

$$\begin{bmatrix} x+y \\ y \\ -3x \end{bmatrix} = \begin{bmatrix} x \\ 0 \\ -3x \end{bmatrix} + \begin{bmatrix} y \\ y \\ 0 \end{bmatrix}$$

2. **Factor out the Parameters:**

$$= x \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} + y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

3. **Identify the Basis:** Every vector in U is a linear combination of $\begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$.

   These two vectors are linearly independent and thus form a basis for U.

$$\text{Basis for } U = \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

4. **Determine the Dimension:** There are **two** vectors in the basis.

Conclusion: The dimension of the subspace U is **2**. Geometrically, this is a plane through the origin in $\mathbb{R}^3$.

1.2 Example 2: A Subspace of $\mathbb{R}^2$ (Geometric Interpretation)

Problem: Find the basis and dimension for the subspace of $\mathbb{R}^2$ of all vectors of the form $\begin{bmatrix} x \\ 3x \end{bmatrix}$.

Solution

Factoring the single parameter gives: $\begin{bmatrix} x \\ 3x \end{bmatrix} = x \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

- **Basis:** $\left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$
- **Dimension:** 1

Conclusion: The dimension is 1. Geometrically, this subspace is a line through the origin in the $\mathbb{R}^2$ plane.

2 The Four Fundamental Subspaces of a Matrix

The starting point for analyzing these subspaces is to find the **Reduced Row Echelon Form (RREF)** of the matrix A. Let's call the RREF matrix B.

2.1 The Null Space, Nul(A)

Basis for the Null Space

The null space is the set of all solutions to $A\mathbf{x} = \mathbf{0}$. Its basis comes from the parametric vector form of the solution.

- **Method:** Solve $B\mathbf{x} = \mathbf{0}$ by expressing basic variables in terms of free variables. Factor out the free variables to get the basis vectors.
- **Dimension (Nullity):** The dimension of Nul(A) is the **number of free variables**.

2.2 The Column Space, Col(A)

Basis for the Column Space

The column space is the span of the columns of A.

- **Method:** Identify the **pivot columns** in the RREF matrix B. The basis for Col(A) consists of the corresponding columns from the **original matrix** A.
- **Dimension (Rank):** The dimension of Col(A) is the **number of pivot columns**.

2.3 The Row Space, Row(A)

Basis for the Row Space

The row space is the span of the rows of A.

- **Method:** The basis for Row(A) is the set of all **non-zero rows** of the RREF matrix B.
- **Dimension:** The dimension of Row(A) is the **number of non-zero rows** in B, which is also the number of pivots.

3 The Rank-Nullity Theorem

The dimensions of these fundamental subspaces are linked by a powerful theorem.

The Rank-Nullity Theorem

For any $m \times n$ matrix A:

1. $\dim(\text{Col } A) = \dim(\text{Row } A)$. This value is the **rank** of A.
2. $\text{rank}(A) + \dim(\text{Nul } A) = n$ (the number of columns).

Another important property is that $\text{rank}(A) = \text{rank}(A^T)$.

Key Intuition

The Rank-Nullity theorem represents a fundamental conservation law for matrices. For a matrix acting on an n-dimensional space, the dimensions are partitioned between the range (the rank) and the kernel (the nullity). The theorem formalizes a crucial trade-off: in dimensionality reduction, every dimension you ”keep” (rank) corresponds to one less dimension of ”lost” information (nullity).

3.1 Example using the Theorem

Problem: A 6×4 matrix A has a rank of 2. Find: a) The dimension of the null space of A. b) The dimension of the row space of A. c) The rank of A^T.

Solution

Here, $m = 6$ (rows), $n = 4$ (columns), and $\text{rank}(A) = 2$.

a) Dimension of Nul(A) (Nullity): Using the theorem: $\text{rank}(A) + \text{nullity}(A) = n$.

$$2 + \dim(\text{Nul } A) = 4 \implies \dim(\text{Nul } A) = 4 - 2 = 2$$

b) Dimension of Row(A): By definition, $\dim(\text{Row } A) = \text{rank}(A)$.

$$\dim(\text{Row } A) = 2$$

c) Rank of A^T: The rank of a matrix is equal to the rank of its transpose.

$$\text{rank}(A^T) = \text{rank}(A) = 2$$

Practical Applications in Data Science & ML

- **Rank (Feature Redundancy):** The rank of a data matrix indicates the number of linearly independent features. A low-rank matrix implies high redundancy (multicollinearity). This is the core idea behind **PCA**, which finds a low-rank approximation of data.
- **Nullity (Model Ambiguity):** If a feature matrix has a non-zero nullity, it means there are infinite solutions for a linear regression model's weights. The null space describes which combinations of features are redundant.
- **Row Space (Feature Importance):** The row space can be seen as the space of all possible "feature profiles." In recommendation systems, if rows represent users, the row space describes the span of all possible user preference patterns.

4 Exercise Problems

1. Find a basis and the dimension for the subspace of $\mathbb{R}^4$ consisting of all vectors of the form $\begin{bmatrix} a-b \\ b+c \\ 2a-c \\ c \end{bmatrix}$.
2. Consider the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 0 & 1 & 1 \end{bmatrix}$.
   a) Find a basis for Nul(A) and its dimension.
   b) Find a basis for Col(A) and its dimension.
   c) Find a basis for Row(A).
   d) Verify that rank(A) + nullity(A) equals the number of columns.
3. A 5×9 matrix A has a null space of dimension 6.
   a) What is the rank of A?
   b) What is the dimension of the row space of A?
   c) Can $A\mathbf{x} = \mathbf{b}$ be consistent for every $\mathbf{b}$ in $\mathbb{R}^5$? (Hint: Does the column space span $\mathbb{R}^5$?)
4. If a 7×5 matrix C has a rank of 5, what can you say about the dimension of Nul(C)? What does this imply about the solutions to $C\mathbf{x} = \mathbf{0}$?